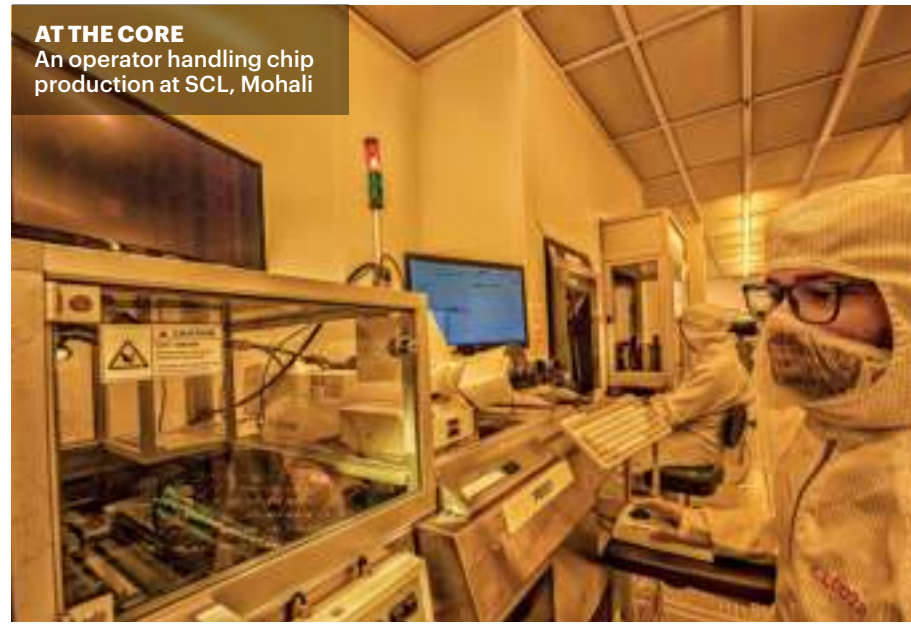
turing if the economy is to thrive."

What Thandavarayan does want the government to do is exploit the significant advantage India has in a large phalanx of chip designers by encouraging global semiconductor companies to set up their R&D operations in India. "Policymakers need to work out how much India can leverage this existing advantage into the very strong ecosystem that we are building," he says. As part of the mission, ISM's Sinha points out, the government has set up a Design Linked Incentive (DLI) scheme with an outlay of Rs 1,000 crore to support start-ups and academic institutions. The initiative provides access to costly Electronic Design Automation (EDA) tools, which an impressive 278 academic institutions and 72 start-ups have availed of.



AT THE CORE
An operator handling chip production at SCL, Mohali

THE BIG PLAYERS

Chip experts give full marks to the Modi government for its judicious curation of private players to build India's semiconductor ecosystem. In its initial approvals, the government targeted the 'low-hanging fruit'—assembly, testing, marking and packaging (ATMP) facilities—which demand far less capital investment than full-fledged fab plants. As a result, of the 10 semiconductor proposals that have been sanctioned so far, only two are for fab units; the remaining eight focus on ATMPs. Each bid was subjected to stringent scrutiny, with the government ensuring that applicants had both financial capacity and credible technology partners.

The biggest approval yet has gone to the Tata Group, which secured a fabrication plant in Dholera, Gujarat, in partnership with Taiwan's Powerchip Semiconductor Manufacturing Corporation (PSMC), the third-largest chipmaker in that country. The total project cost is estimated to be Rs 91,000 crore, of which 50 per cent will come from central subsidy and 20 per cent from Gujarat; Tata's contribution will be Rs 27,000 crore. Tata has also been sanctioned an assembly and testing facility in Morigaon, Assam, in collaboration with Bosch as technology partner. This project entails an investment of Rs 27,000 crore, with the Centre and the Assam government bearing 70 per cent of the cost.

> "To address the shortage of skilled manpower, the government needs to establish centres of excellence in collaboration with academic and global semiconductor companies"
>
> **KATHIR THANDAVARAYAN**
> *Partner, Deloitte South Asia*



Meanwhile, US semiconductor major Micron is building a colossal Rs 22,516 crore ATMP plant on a 75-acre campus in Sanand. Once fully operational, it will process up to 1.35 billion memory chips annually. Micron will also receive 70 per cent of its project cost as government grants. Mysuru-based Kaynes Semicon tied up with UST Global, US, for a Rs 3,300 crore state-of-the-art OSAT facility in Sanand with near similar incentive terms as Micron.

The largesse is being extended beyond Gujarat. In Uttar Pradesh's Jewar area, where an international airport becomes operational this month, HCL is building a Rs 3,700 crore display driver chip unit in partnership with Foxconn, the world's largest electronics contract manufacturer. In Odisha, US-based 3D Glass Solutions is setting up a Rs 1,943 crore facility for advanced packaging and glass substrate components and Chennai-based SicSem, in partnership with UK's Clas-SIC Wafer Fab, is establishing a semiconductor fab with Rs 2,066 crore investment. Both these facilities are dedicated mainly to the defence and automotive